

Unmanned Aircraft Systems UAS

Operations

FAA Remote Pilot Part 107 Exam Prep-Regulations



FAA Remote Pilot Test Prep

Topics Covered in this lesson

- Introduction
- Communication Procedures
- sUAS Frequencies
- Emergency Procedures
- Lost link
- Flight termination
- Flyaways
- Loss of GPS
- Battery Fires
- Aeronautical Decision Making
- Physiology
- Maintenance and Inspection Procedures

Operations Introduction

- ◆ The remote PIC of an sUAS is responsible for determining whether that aircraft is in condition for safe flight.
- ◆ The remote PIC must inspect the sUAS before each flight.
- ◆ Preflight inspection should be a thorough and systematic means by which the remote PIC determines that the aircraft is in a safe condition for flight.
- ◆ When in doubt, remove and replace any damaged part.
- ◆ Always consult with the manufacturer for the proper procedure for the repair and/or replacement of parts.

Operations Introduction



The remote PIC must determine the aircraft is in a condition for safe flight.



Preflight inspection must be thorough, systematic, and completed before each operation.

Sample Preflight Inspection Checklist

- | | |
|--|---|
| ☒ Visually inspect system components | ☒ Inspect registration markings |
| ☒ Inspect airframe structure & control surfaces | ☒ Verify adequate energy supply |
| ☒ Check fuel/battery for correct type & quantity | ☒ Inspect avionics and antenna(s) |
| ☒ Securely attached camera/equipment | ☒ Calibrate UAS compass |

Operations Communications Procedures



The FAA requires that the remote PIC and other crew members coordinate to:

- Scan the airspace in the operational area any potential collision hazard.
- ◀ □ Visual lined line of-sight shotos sirect airw and clisconedt.
- Maintain awareness of the position PIC through direct visual observation.



If communication with ATC is required, the remote PIC should use standard communication phtraseology.

Phonetic Alphabet

A -> Alpha

B -> Bravo

C -> Charlie

D -> Delta

E -> Echo

F -> Fox-trot

G -> Golf

H -> Hotel

J -> India

J -> Juliet

K -> Kilo

L -> Lima

M -> Mike

N -> November

O -> Oscar

P -> Papa

Q -> Quebec

R -> Romeo

S -> Sierra

T -> Tango

U -> Uniform

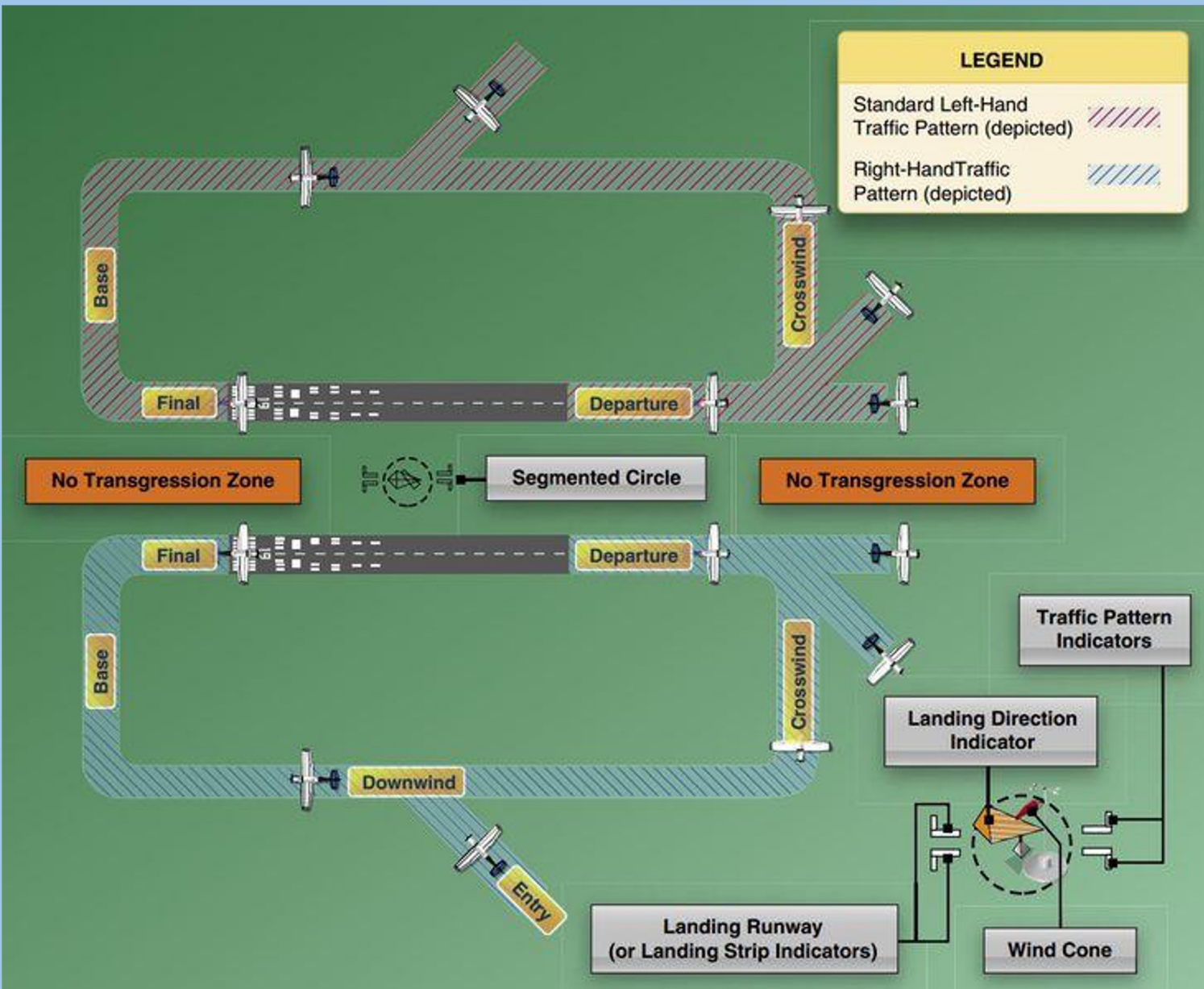
V -> Victor

W -> Whiskey

X -> X-ray

Y -> Yankee

Z -> Zulu



“On the 45 for 18”

“Entering a left downwind for 18”

“Turning left base runway 36”

“Final 36”

“Turning final 36”

“Short final 36”

“turning left crosswind 18”

“upwind 36 departing the area to the NW”

“departing runway 18 will remain in the pattern”

“Departing 36 close traffic”

“You are operating your sUAS:

1 mile north of the airport

$\frac{1}{4}$ mile east of the airport

2 miles north west of the airport.

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Communication Procedures

The remote PIC and all crew members should be familiar with common communication frequencies and when radio calls and position reports are expected.

Facility at Airport	Frequency Use	Communication/Broadcast Procedures		
		Outbound	Inbound	Practice Instrument Approach
UNICOM (no tower or FSS)	Communicate with UNICOM station on published CTAF frequency (122.7, 122.8, 122.725, 122.975, or 123.0). If unable to contact UNICOM station, use self-announce procedures on CTAF.	Before taxiing and before taxiing on the runway for departure.	10 miles out. Entering downwind, base, and final. Leaving the runway.	
No tower, FSS, or UNICOM	Self-announce on MULTICOM frequency 122.9.	Before taxiing and before taxiing on the runway for departure.	10 miles out. Entering downwind, base, and final. Leaving the runway.	Departing final approach fix (name) or on final approach segment inbound.
No tower in operation, FSS open	Communicate with FSS on CTAF frequency.	Before taxiing and before taxiing on the runway for departure.	10 miles out. Entering downwind, base, and final. Leaving the runway.	Approach completed/terminated.
FSS closed (no tower)	Self-announce on CTAF.	Before taxiing and before taxiing on the runway for departure.	10 miles out. Entering downwind, base, and final. Leaving the runway.	
Tower or FSS not in operation	Self-announce on CTAF.	Before taxiing and before taxiing on the runway for departure.	10 miles out. Entering downwind, base, and final. Leaving the runway.	

Operations Communication Procedures

- Aircraft call signs should be spoken in their entirety, using the phonetic alphabet.
- Numbers are spoken individually.
- The remote PIC and all crew members should understand common phrases used in air traffic radio communication.

"Acknowledge"	Let me know that you have received my message.
"Advise intentions"	Tell me what you plan to do.
"Affirmative"	Yes.
"Final"	Commonly used to mean that an aircraft is on the final approach course or is aligned with a landing area.
"Hold for"	(takeoff clearance, release, landing/taxiing aircraft, etc.) Stay in place where you are currently located.
"Hold short"	November 477ZA, runway four, taxi via Echo, hold short runway two five at taxiway Delta.
"How do you hear me?"	A question relating to the quality of the transmission or to determine how well the transmission is being received.
"Immediately"	Used by ATC or pilots when such action compliance is required to avoid an imminent situation.
"Line up and wait"	Used by ATC to inform a pilot to taxi onto the departure runway in takeoff position and LINE UP and WAIT. <i>It is not authorization for takeoff.</i> It is used when takeoff clearance cannot immediately be issued because of traffic or other reasons.
"Negative"	"No" or "permission not granted" or "that is not correct."
"Read back"	Repeat my message back to me.
"Roger"	I have received all of your last transmission. <i>Should not be used to answer a question requiring a yes or no answer.</i>
"Stand by"	Means the controller or pilot must pause for a few seconds, usually to attend to other duties of a higher priority. Also means to wait, as in "stand by for clearance." The caller should reestablish contact if a delay is lengthy. "Stand by" is not an approval or denial.
"Unable"	Indicates inability to comply with a specific instruction, request, or clearance.
"Verify"	Request confirmation of information (for example, "verify assigned altitude").
"WILCO"	I have received your message, understand it, and will comply with it.
"Without delay"	With a sense of urgency, proceed with approved instructions in a rapid manner.

Operations Communication Procedures

The term “initial radio contact” or “initial callup” means the first radio call you make to a given facility, or the first call to a different controller or flight service specialist within a facility. Use the following format:

- ▣ Name of the facility being called.
- ▣ Your full aircraft identification as filed in the flight plan.
- ▣ Type of message to follow or your request if it is short.
- ▣ The word “over” if required. This is typically omitted and is no longer commonly used.

Sample Communication Phraseology

What does all of this mean?

- ◆ Aircraft- “Bozeman approach skyhawk niner two lima Charlie”
- ◆ ATC “Skyhawk niner two lima Charlie say request”
- ◆ Aircraft “We would like VFR flight following to Sierra Delta Yankee, Sidney-Richland. Currently 6 thousand three hundred climbing seven thousand five hundred niner two lima charlie”
- ◆ ATC “niner two lima Charlie squawk 3461 and ident”
- ◆ Aircraft “3461 niner two lima Charlie”
- ◆ ATC “niner two lima charlie radar contact 5 mile SE Bozeman field, climb maintain seven thousand five hundred, maintain VFR, Bozeman altimeter two niner zero niner”
- ◆ Aircraft “climb maintain seven thousand five hundred, two niner zero niner Skyhawk niner two lima Charlie”

Operations

sUAS Frequencies

- ◆ Radio frequency bands are regulated by the Federal Communications Commission (FCC).
- ◆ Most common sUAS frequencies are 2.4GHz and 5.8 GHz.
- ◆ These frequencies are close to wireless network frequencies and can cause interference.
 - Frequency interference increases fly away risk near densely populated areas.
- ◆ 5.8GHz used for control.
- ◆ 2.4 GHz used to transmit photos and videos.
- ◆ Consult the FCC for frequency licensing info at

www.fcc.gov/licencing-databases/licencing

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Emergency Procedures

- ◆ **Lost Link** - Loss of communications link between PIC and aircraft. Notify ATC if in controlled airspace.
- ◆ **Flight Termination** – Intentional process of controlled flight into the ground to terminate a flight.
- ◆ **Flyaways** – Typically occurs after a loss link. Aircraft flies uncontrolled and unpredictably. If in ATC controlled airspace, notify ATC immediately.
- ◆ **Loss of GPS** – Pilots should not become over reliant on GPS. Remote PIC should have plan in place if loss of GPS occurs. Loss of GPS is not typically an emergency.
- ◆ **Battery Fires** – Lithium metal and lithium-ion batteries are highly flammable, and subject to thermal runaway. Pose a significant hazard to sUAS. Never use damaged batteries.

Operations

Aeronautical Decision Making

Develop a culture of safety

- ◆ **Situational awareness** – obtain as much information as possible before a flight. Knowledge of airspace, weather, privacy, and ATC requirements all contribute to a remote PICs situational awareness.
- ◆ **Risk-based Aeronautical Decision Making (ADM)** – a systematic approach to the mental process used by pilots to consistently determine the best course of action in response to a given set of circumstances.
- ◆ **Crew Resource Management (CRM)** – the effective use of all available resources, human, hardware, information, etc., prior to, and during flight, to ensure a successful outcome of the operation.
- ◆ **Safety Management Systems (SMS)** – the formal, top-down business-like approach to managing safety risk, which includes a systemic approach to managing safety, including the necessary organizational structures, accountabilities, policies and procedures.

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Aeronautical Decision Making

The Five Hazardous Attitudes

Anti-Authority: “Don’t tell me.”

This attitude is found in people who do not like anyone telling them what to do. In a sense, they are saying, “No one can tell me what to do.” They may be resentful of having someone tell them what to do, or may regard rules, regulations, and procedures as silly or unnecessary. However, it is always your prerogative to question authority if you feel it is in error.

Impulsivity: “Do it quickly.”

This is the attitude of people who frequently feel the need to do something, anything, immediately. They do not stop to think about what they are about to do; they do not select the best alternative, and they do the first thing that comes to mind.

Invulnerability: “It won’t happen to me.”

Many people falsely believe that accidents happen to others, but never to them. They know accidents can happen, and they know that anyone can be affected. However, they never really feel or believe that they will be personally involved. Pilots who think this way are more likely to take chances and increase risk.

Macho: “I can do it.”

Pilots who are always trying to prove that they are better than anyone else think, “I can do it—I’ll show them.” Pilots with this type of attitude will try to prove themselves by taking risks in order to impress others. While this pattern is thought to be a male characteristic, women are equally susceptible.

Resignation: “What’s the use?”

Pilots who think, “What’s the use?” do not see themselves as being able to make a great deal of difference in what happens to them. When things go well, the pilot is apt to think that it is good luck. When things go badly, the pilot may feel that someone is out to get me, or attribute it to bad luck. The pilot will leave the action to others, for better or worse. Sometimes, such pilots will even go along with unreasonable requests just to be a “nice guy.”

Operations Physiology

- ♦ **Hyperventilation** – a deficiency of carbon dioxide within the body. Can be the result of rapid breathing from stress, anxiety or fear.
- ♦ **Dehydration** - Critical loss of water from the body. (headaches, fatigue, cramps, sleepiness, and dizziness are symptoms).
- ♦ **Heatstroke** – Any inability of the body to regulate temperature. (symptoms common to dehydration symptoms but can also become complete collapse of victim).
- ♦ **Hypothermia** – excessively low body temperature. (shivering, clumsiness, slurred speech, confusion, low energy, discoloration of the skin (red or blue) loss of consciousness).
- ♦ **Stress** – Causes by many factors including working with new or inexperienced crew members, having spectators or government officials on hand etc. Prior proper planning can help reduce stress.
- ♦ **Fatigue** – Acute (short term) and chronic (prolonged) tiredness. Fatigue is a factor in many aviation accidents. Fatigue can be caused by lack of sleep, stress, length of time spent on a specific task etc. Chronic fatigue can be caused when acute fatigue has not been fully recovered from.
- ♦ **Chronic Stress** – Long term untreated stress. Best way to treat stress is to remove or limit stressors. Chronic stress can weaken the immune system, cause high blood pressure, anxiety, confusion, mental errors, insomnia.

Operations

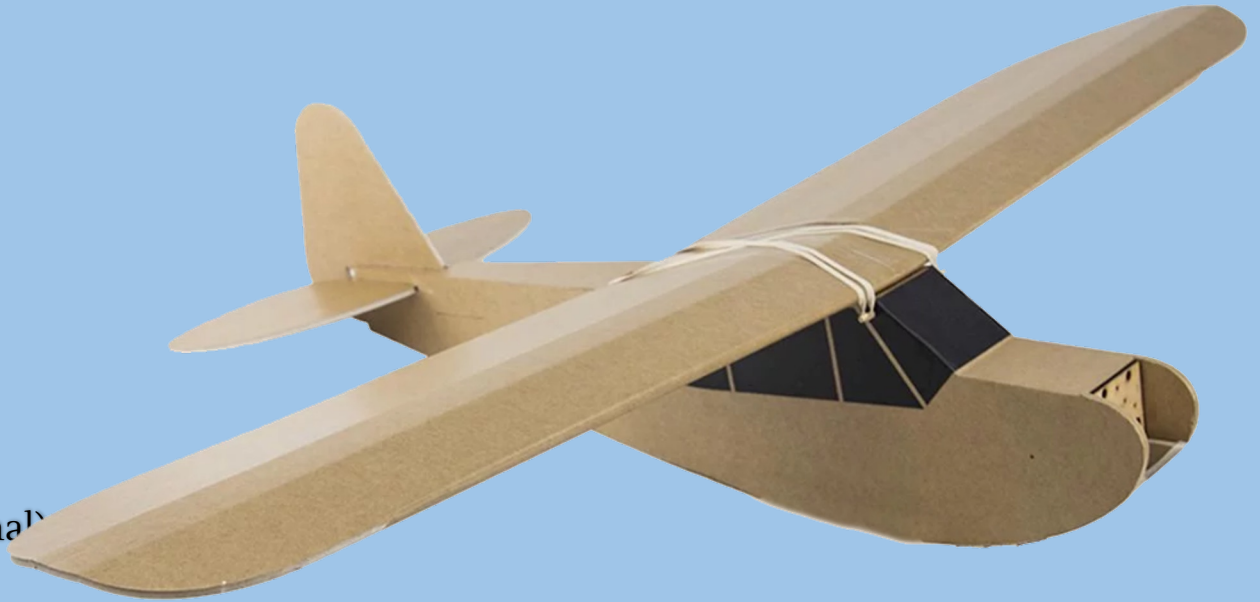
Maintenance and Inspection Procedures

- ◆ Follow all manufacturer maintenance recommendations to achieve the longest and safest service life of the sUAS.
- ◆ If no procedures exist, it is recommended that the remote PIC develop his/her own procedures.
- ◆ Follow the most restrictive limitations or component cycle life in order error on the side of safety.

It is the remote PIC's responsibility to make sure the UAS (as well as the remoter pilot) is in a safe condition for flight.

What does a training flight look like?

- ◆ Check weather
- ◆ Establish a flight zone.
- ◆ Establish an observer area.
- ◆ Establish a pilot and trainer area.
- ◆ Pre-flight aircraft
 - ▢ Flight controls free and correct
 - ▢ Battery secure
 - ▢ Dual control check
 - ▢ Visual inspection of aircraft
 - ▢ Check airspace (visually and on VFR sectional)
- ◆ Preflight student
 - ▢ alert
 - ▢ distractions removed/mitigated
 - ▢ Understand controls/what will happen next



References

- ◆ Marshall, D.; Barnhart, R., Shappee, E., Most, M. (2016). *Introduction to Unmanned Aircraft Systems*. New York, NY: CRC Press.